MIL-I-38135B(USAF)
11 March 1971
SUPERSEDING
MIL-I-38135A(USAF)
5 August 1963

MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED AVU-19/A, STANDBY

1. SCOPE

1.1 This specification covers the requirements for a standby integrally lighted 1-7/8 inch round-dial pitot-static airspeed indicator, designated AVU-19/A.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this speicfication to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

FSC 6610

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specification, Order of Precedence for the Selection of
MIL-STD-471	Maintainability Demonstration
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS28105	Window, Dial-Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33638	Case, Instrument, Flange Mounted, Aircraft
MS33649	Bosses, Fluid Connection - Internal Straight Thread

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Materials shall conform to applicable specifications and as specified herein. When materials are used that are not specifically designated, they shall be of the best quality, of the highest practicable weight, and shall be suitable for the purpose intended.

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3.3.2 Fungus-proof-materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a heremetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

* 3.3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.3.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

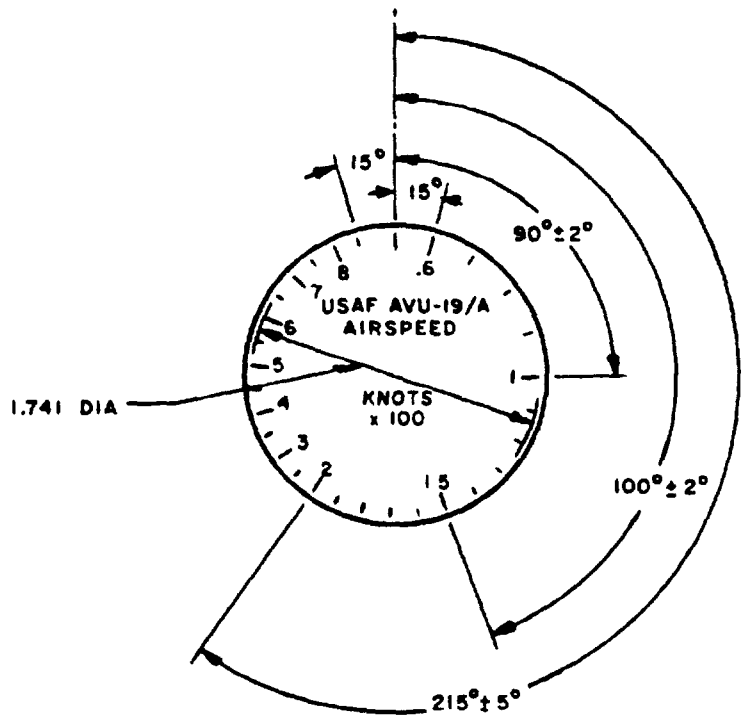
3.4.1 Design. The indicator shall be designed to provide an indication of indicated airspeed as a function of the difference between pitot and static pressures. The indicator shall conform to figures 1 and 2.

3.4.1.1 The design shall be reasonably simple to permit overhaul or repair without the use of special tools or fixtures.

3.4.2 Construction. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

* 3.4.3 Reliability. The indicator shall have a minimum acceptable mean-time-between failures (MTBF) of at least 1,200 hours to a 90 percent confidence factor.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED,
TOLERANCES DECIMALS ± 010
DEGREES ± 1 0

Figure 1. Dial

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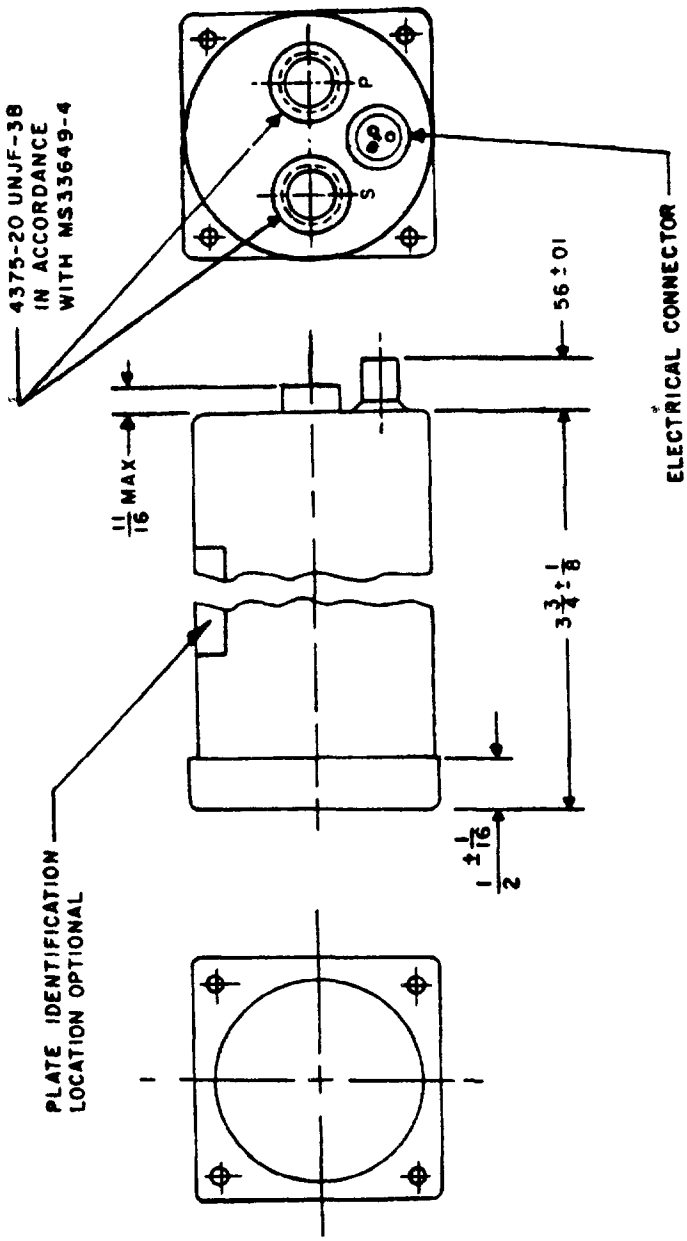


Figure 2. Case

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* 3.4.5 Maintainability program. The contractor shall establish a maintainability program which shall include but not be limited to: Defining levels (line, shop, and depot) of support required to maintain the indicator, predicting expected failure modes, and defining equipment required to repair any failed components.

* 3.4.5.1 Maintainability. The following maintainability requirements shall be considered in the design of the indicator:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly or component

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

3.5 Performance. The indicator shall meet the requirements specified herein under the following conditions:

a. Temperature: Temperatures ranging from -65° to $+71^{\circ}$C

b. Vibration

(1) Vibration error: Vibration at an amplitude of 0.003 to 0 005 inch at a frequency of 500 to 3,000 cpm

(2) Vibration failure: Vibration at an amplitude of 0.018 to 0.020 inch at frequencies of 1,000, 2,000, and 3,000 cpm for a period of 3 hours

c. Salt fog: Exposure to simulated salt sea atmosphere for a period of 50 hours

d. Humidity: Relative humidity up to 100 percent

e. Fungus resistance: Exposure to fungus growth as encountered in tropical climates

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f. Acceleration: Applied acceleration forces up to 20g

g. Range: Continuous without change in calibration within a range of 60 to 850 knots

h. Overpressure: Overpressures as listed in table I

i. Sand and dust: Exposure to sand and dust as encountered in desert areas

j. Rain: Exposure to simulated rainfall.

3.5.1 Magnetic effect. The indicator shall not cause a free compass to deflect more than 1° when revolved about a short-bar magnetic compass with a field intensity of 0.17 to 0.19 oersted.

3.5.2 Case leakage. Case leakage shall not exceed 0.01 inch Hg for a 1-minute period under the conditions specified in section 4.

3.5.3 Diaphragm capsule leakage. Diaphragm leakage shall not cause more than 1-knot deflection of the pointer for a 1-minute period under the conditions specified in section 4.

3.5.4 Scale, friction, and position errors. The scale, friction, and position errors shall not exceed the tolerances specified in tables I and II.

* 3.5.5 Dielectric strength. The indicator shall withstand application of 500V for a period of 5 seconds.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.7 Case. The indicator case shall conform to MS33638 for the 2-inch nominal size except as shown on figure 2. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in flat black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental conditions specified herein. The body of the case shall be made of aluminum alloy which shall be uniform in texture and shall have a smooth surface.

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TABLE I Test Tolerances

Test	± Tolerance
Friction	
60 to 200 knots	3 knots
200 to 850 knots	10 knots
Position error	
60 to 200 knots	3 knots
200 to 850 knots	10 knots
Pointer zero position	10 knots
Vibration error:	
Test airspeed 200 knots	
Maximum pointer vibration	5 knots
Overpressure	
Static connection 10 inches H_2O	
Pitot connection 73 inches Hg	
Test airspeed 200 knots	5 knots
Vibration failure	
Test airspeed 200 knots	5 knots
Low temperature operation	5 knots
High temperature operation	5 knots
Damping	
Time required for the airspeed pointer to travel from full scale to 120-knot graduation	$1\ ^{+1.0}_{-0.2}$ second
Shelf life	3 knots between 60 and 200 knots and 10 knots between 200 and 850 knots

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TABLE II. Scale Error at Room Temperature

Test Points Speed (Knots)	Differential Pressure (Inches Hg)	Tolerance (Knots)
60	0.172	±3
80	0.308	±3
100	0.482	±3
120	0.696	±3
150	1.092	±3
200	1.962	±5
300	4.541	±10
400	8.397	±15
500	13.797	±20
600	21.109	±20
700	30.816	±20
800	43.009	±20

3.7.1 Bezel ring. The bezel ring shall be made of aluminum alloy and shall be held in place by means of screws properly secured by lock washers.

3.7.2 Cover glass. The cover glass shall conform to MS28105-7 and shall be properly sealed into the bezel ring. The bezel shall be held in place by means of screws properly secured by lockwashers or similar devices of approved design. If the cover glass is used as part of the lighting system, a glass may be used with characteristics deviating from the glass specified herein to the extent required for the lighting function provided it does not interfere with the readability of the instrument.

3.7.3 Dial. The dial shall be fastened securely into the case or upon the frame of the mechanism by at least two screws in such manner that it will not loosen or turn when the indicator is vibrated. The dial shall conform to figure 1.

3.7.4 Dial and cover glass location. The distance between the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the inside surface of the cover glass and the dial shall not exceed 0.125 inch.

* 3.7.5 Cover glass and lighting wedge. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806 in addition to withstanding the environmental conditions specified herein.

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3.8 Pointer. The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. It shall be firmly attached to the mechanism and shall be readily adjustable. The pointer shall conform to MS33585-8 and shall be finished in dull white, color No. 37875 of FED-STD-595.

3.8.1 Limitation of pointer movement. The airspeed pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 358° of arc clockwise from the no-pressure position when the indicator is subjected to overpressure tests. The design of the stops shall be subject to the approval of the procuring activity.

* 3.9 Electrical connector. The electrical connector shall be a Deutsch type DM5606-3P, or equal, and shall be located as shown on figure 2.

3.10 Integral lighting. The indicator shall be integrally lighted with both red and white lighting circuits as shown on figure 3. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. The lighting shall operate on 5V ac. Lighting acceptability of production indicator shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.10.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured in foot-lamberts for brightness level. A minimum of 20 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded overall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 500 hours of operation to determine any change in brightness or color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action.

3.11 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.11.1 Capillary tubing. A sufficient length of approved capillary tubing having a minimum inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the instrument case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the requirements of the damping test specified herein.

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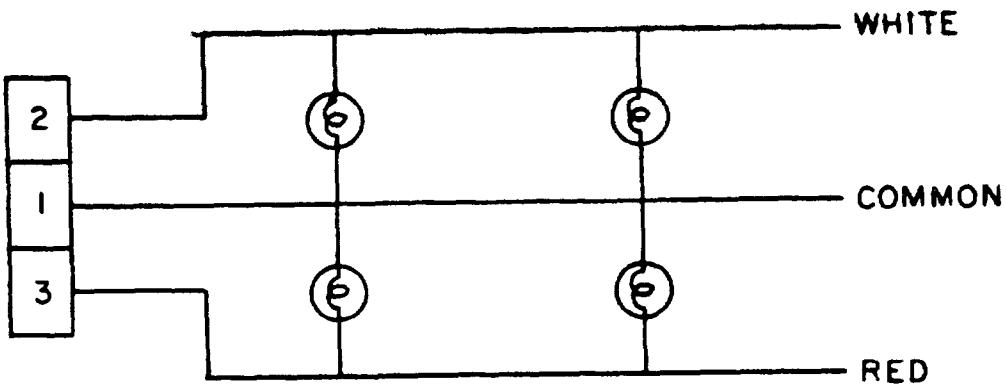


Figure 3. Wiring Diagram

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3.11.2 Standard pressure. The standard pressure used in calibrating airspeed indicators for use with pitot-static tubes shall be in accordance with the values specified in table II.

3.11.3 Plug. Each static and pitot pressure connection shall be provided with a vented plug of a material and design acceptable to the procuring activity. The vent hole shall be 1/16 ±1/64 inch in diameter.

3.12 Markings. All markings shall be so durable as to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to MS33558 Numerals shall distinctly indicate the graduations to which each applies.

3.12.1 Finish. Dial markings "USAF AVU-19/A" and "AIRSPEED" shall be 0.047 ±0.0156 inch in height and shall be permanently and legibly marked on the dial. These markings, and all other markings not otherwise specified, shall be finished in lusterless black, color No. 37038 of FED-STD-595. The following dial markings shall be finished in lusterless white, color No. 37875 of FED-STD-595. The dimensions shall be as indicated.

	Height or Length Inch ±0.016	Width of Line or Graduation Inch ±0.005
Numerals .6, 1, 1.5, 2, 3, 4, 5, 6, 7, 8	0 140	0.020
100-knot graduations	0.171	0.020
50-knot graduation	0.156	0.020
10-knot graduations	0.140	0.015
Lettering KNOTS x 100	0.093	0.020

3.12.2 Visibility of dial markings. At least 1/16 inch of graduations, numerals, pointers, and other markings on the dial shall be visible from any point within the frustum of a cone, the side of which makes an angle of not less than 30° with the perpendicular to the dial and the small diameter of which is the aperture of the dial.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.14 Weight. The weight of the indicator shall not exceed 1.25 pounds.

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3.15 Finishes and protective coatings

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.15.1.1 Dials, small holes, and case inserts need not be anodized.

3.15.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.15.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.4 Coatings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.15.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.15.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance withe QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

.17 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.17.2 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

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3.17.3 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it shall be understood that the test is to be made at atmoshpheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.3.4 Vibration. A vibration stand shall be used that will vibrate at any desired frequency between 500 and 3,000 cpm and will subject the indicator to such vibration that a point on the case will describe, in a plane inclined 45° to the horizontal plane, a circle of the size diameter specified herein.

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* 4.4 Preproduction testing

* 4.4.1 Test samples. The test samples shall consist of nine indicators representative of the production equipment. Three indicators shall be subjected to all the tests except reliability and the other six shall be subjected to reliability only. The indicators shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831(see 6.2.1).

* 4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Case leakage

c. Diaphragm capsule leakage

d. Scale error at room temperature

e. Friction

f. Individual tests of MIL-L-25467 and MIL-L-27160.

4.5.2 Sampling plans and tests

* 4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order, and subjected to the following tests as described under 4.6:

a. Individual tests

b. Position error

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c. Low temperature operation

d. High temperature operation

e. Vibration error

f. Dielectric strength

g. Sampling plan A tests of MIL-L-25467 and MIL-L-27160.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 indicators shall be selected at random from the first 15 indicators on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Low temperature exposure

c. High temperature exposure

d. Acceleration

e. Humidity

f. Salt fog

g. Vibration failure

h. Fungus

i. Damping

j. Magnetic effect

k. Pointer zero position error

l. Overpressure

m. Dust

n. Rain

o. Sampling plan B tests of MIL-L-25467 and MIL-L-27160.

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* 4.5.2.3 Sampling plan C. Unless otherwise specified (see 6.2), six indicators shall be selected at random from each 200 or less produced on the contract and subjected to the following tests as described under 4.6. Reliability accept-reject criteria shall be as specified in 4.6.25.1.

a. Individual tests

b. Reliability test.

* 4.5.2.4 Sampling plan D. Unless otherwise specified (see 6.2), one indicator shall be selected at random from each monthly production quantity and subjected to the shelf life test specified in 4.6.26.

4.5 2.5 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.5.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and dimensions.

4.6.2 Case leakage. The pitot and static pressure connections of the indicator shall be joined with a Y connection and connected to the same mercury manometer and a source of vacuum and pressure. A suction of 25 inches Hg shall be applied to the pitot and static connections of the indicator. With the source disconnected for a period of 1 minute, the difference in the mercury levels shall not change by more than 0.01 inch. This procedure shall be repeated with a pressure of 10 inches Hg. During a period of 1 minute, the mercury levels shall not change more than 0.01 inch Hg.

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4.6.3 Diaphragm capsule leakage. With the static connection open to the atmosphere, a pressure sufficient to produce approximately full-scale deflection of the pointer shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of 1 minute, the pointer shall not change its position by more than 1 knot.

4.6.4 Scale error at room temperature. The indicator shall be tested for scale errors (see 6.3.1) at the test points shown in table II. The test shall be made by subjecting the indicator to the pressures specified to produce these readings, first with increasing pressures, then with decreasing pressures. For increasing pressures, the pressure shall be brought up to but not exceeding the pressures required to give the desired reading and for decreasing pressures, the pressure shall be brought down to but not below the pressures required to give the desired reading. The errors shall not exceed those specified in table II.

4.6.5 Friction. The indicator shall be tested for friction at the 60-, 100-, 150-, 200-, 300-, 400-, 500-, 600-, 700-, and 800-knot indications. The pressure shall be increased to bring the pointer to the approximate desired reading and held constant while two readings are taken, the first before the indicator is tapped, the second after the indicator is tapped. The difference between any two readings shall not exceed the values specified in table I. The pointer shall move smoothly while the pressure is varied uniformly without vibrating the indicator. This test may be combined with the scale error at room temperature test.

4.6.6 Position error. While the indicator is held in any one position and being tapped, the indicator reading shall not differ from its reading when held in any other position by more than the tolerances specified in table I.

4.6.7 Low temperature operation. The indicator shall be subjected to a temperature of $-54^{\circ} \pm 2^{\circ}$ C for a period of 4 hours. At the end of this period and while still at this temperature, the indicator shall be tested for scale error. The errors at the test points shall not exceed those specified in table II by more than the value specified in table I. No damage to any part of the indicator shall result from this test.

4.6.8 High temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of $71^{\circ} \pm 2^{\circ}$ C for a period of 4 hours. At the end of the 4-hour period and while still at the high temperature, the indicator shall be tested for scale errors at room temperature. The errors at the test points shall not exceed those specified in table II by more than the value specified in table I. No damage to any part of the indicator shall result from this test.

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4.6.9 Vibration error. The indicator shall be mounted on the vibration stand in its normal operating position and secured by flange screws positioned as on an aircraft panel. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm, so that any point on the indicator case will describe the circular path of amplitude specified and the circular path will be inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the airspeed indicator shall not occur in this range. While the indicator is being vibrated, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed the tolerance specified in table I.

* 4.6.10 Dielectric stength. A voltage of 500V shall be applied from terminals 1-2-3 (together) to ground for a minimum of 5 seconds. There shall be no insulation breakdown nor permanent damage to the indicator as a result of this test. The voltage shall be raised from zero to maximum by rheostat control and not by switching.

* 4.6.11 Low temperature exposure. The indicator shall be subjected to a low temperature exposure test in accordance with MIL-STD-810, method 502, for a 48-hour period at -65° ±2°C. At the end of the 48-hour period and while still at the low temperature, the indicator shall be tested for scale error as specified in 4.6.4. The errors at the test points shall not exceed those specified in table II by more than the values specified in table I. After the indicator has returned to room temperature, it shall meet the individual tests.

* 4.6.12 High temperature exposure. The indicator shall be subjected to a high temperature exposure test in accordance with MIL-STD-810, method 501, procedure I, except for a period of 24 hours. After the indicator has returned to room temperature, it shall meet the individual tests.

4.6.13 Acceleration. The indicator shall be mounted in its normal operating position on a rotating arm with the rear of the case toward the center of rotation and subjected to an acceleration of 20g for a period of 30 seconds. This test shall be repeated with the indicator mounted successively in the following positions: (1) rotated 90° in azimuth with respect to the original position and (2) dial horizontal, with the zero graduation toward the center of rotation. At the end of this acceleration, the indicator shall meet the individual tests. No damage to the indicator shall result from this test.

* 4.6.14 Humidity. The indicator shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I.

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* 4.6.15 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. At the end of the 50-hour period, there shall be no damage to any part of the indicator that would affect the performance of the instrument. The indicator shall then be subjected to and shall meet the individual tests.

4.6.16 Vibration failure. This test shall be conducted using the test differential pressure specified in table I as the test point. The indicator indication at the test differential pressure specified shall be determined with pressures increasing. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 1-hour period at a frequency fo 1,000 cpm, followed by a 1-hour period at a frequency of 2,000 cpm, and then by a 1-hour period at a frequency of 3,000 cpm. At the end of the vibration period, the indicator shall be subjected to the pressure specified to produce the test point reading (increasing pressure) The difference between this indication and that determined prior to the vibration for the same differential pressure shall not exceed the values specified in table I. No looseness in the mechanism or damage to any part of the indicator shall result from this test

* 4.6 17 Fungus. The indicator, including applicable external connectors, shall be subjected to a fungus test in accordance with MIL-STD-810, method 508. There shall be no deterioration that would adversely affect subsequent operation.

4 6.18 Damping. Sufficient pressure shall be applied to the pitot connector to produce full-scale deflection of the airspeed pointer. The pressure shall suddenly be released. The time required for the airspeed pointer to reach the value specified in table I shall be equal to the time and within the tolerance specified therein.

4 6.19 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from the center of a short-bar magnetic compass with compensator magnets removed and which is set up in a uniform magnetic field having a horizontal intensity between 0.17 to 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. The maximum deflection shall not exceed 1°.

4.6.20 Pointer zero position error. The zero position of the pointer with no differential pressure applied to the indicator shall be within the values specified in table I.

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4.6.21 Overpressure. The pressure specified in table I shall be applied to the pitot pressure and static connections of the indicator for a period of 10 minutes. Not less than 5 minutes following the application of overpressure, the indicator shall be tested for scale error as specified in the scale error at room temperature test. The errors at the test points shall not exceed those specified in table II by more than the value specified in table I.

* 4.6.22 Dust. The indicator shall be subjected to a dust test in accordance with method 510, procedure I of MIL-STD-810.

* 4.6.23 Rain. The indicator shall be subjected to a rain test in accordance with method 506, procedure I of MIL-STD-810.

4.6.24 Lighting. Indicator lighting shall be tested in accordance with MIL-L-25467 and MIL-I-27160.

* 4.6.25 Reliability. A minimum of six indicators shall be subjected to a reliability test in accordance with the following:

a. Each indicatorshall be subjected to a pressure cycling of a minimum of 3 cycles per hour for a minimum of 880 hours or 2,640 cycles consisting of a pressure variation equivalent to approximately 3/4 of full scale. Each pressure cycle shall consist of operation to maximum required pressure and return to ambient. The maximum scale value shall be maintained for not less than 2 minutes. Approximately 2 minutes at ambient pressure shall be allowed between each pressure cycle. During the fifth cycling period, the indicator shall be vibrated at an amplitude of 0.009 to 0.011 inch and a frequency of 2,100 cpm. The cycling period, number of hours or cycles, and cycling temperature shall be accomplished in accordance with table III. After each period of cycling, the indicator shall be subjected to and shall meet the scale error at room temperature test, except that only six test stations, representative of the low range, midrange, and high range, shall be tested. The room temperature tolerances specified in table II shall apply. During each scale error test, when any two test points exceed allowable tolerances by 2 knots, one malfunction shall be charged against this test. Upon completion of the final cycle, the indicator shall be rested at ambient pressure for a period of 48 hours. The indicator shall then be subjected to and shall meet the individual tests. During the individual tests, failure to meet any test shall be considered a malfunction. For each five test points out of tolerance during the scale error test, one malfunction shall be charged.

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b. In case of malfunction of any kind, the malfunction shall be recorded and the reliability test interrupted while repairs are made. The test shall then be continued from the point of interruption. If recalibration is required to pass the scale error test, an additional malfunction shall be charged against this test.

TABLE III. Reliability Cycling

Cycling period	Hours or Cycles		Ambient Temperature °C
1	150	450	25
2	100	300	55
3	150	450	25
4	100	300	-35
5	100	300	25
6	100	300	55
7	100	300	-35
8	100	300	25

* 4.6.25.1 The test shall demonstrate an MTBF of 1,200 hours to a 90 percent confidence level. This level of reliability shall be verified when the six indicators have satisfactorily complied with the requirements of table IV.

* TABLE IV. Reliability Accept-Reject Criteria

Preproduction		Production	
Total Allowable Failures	T/1200	Total Allowable Failures	T/1200
2	6.0	0	1.6
3	6.8	1	3.3
4	7.6	2	5.0
5	8.5	3	6.0
		4	7.0
		5	8.0

* 4.6.26 Shelf life. The indicator shall be placed in storage for a period of not less than 30 days. During the 30-day period, the indicator shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal abmitent pressure vibrations. The temperature shall remain essentially constant; however, normal room temperature variations will be permissible. After the 30-day period, the indicators shall again be subjected to and shall meet the individual tests. In addition to meeting the tolerances specified for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table I.

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* 4.6.27 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471 and shall be performed on a minimum of three selected repairs. The tasks shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, indicators shall be individually preserved and packaged in accordance with MIL-P-116, method IIC.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, or physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance wtih MIL-STD-1186.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection. Cusioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cusioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

INDICATOR, INDICATED AIRSPEED AVU-19/A

6. NOTES

6.1 Intended use. The AVU-19/A indicator covered by this specification is intended for use in aircraft as a standby instrument to indicate the speed relative to the air at sea level and the equivalent seal level speed corresponding to the actual sustaining force of the air at any other altitude.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B, C, and D tests will not be conducted

c. Levels of packaging and packing required.

6.2.1 Test report. The contractor will submit three copies of the test report specified in 4.4.2 to the procuring activity for approval.

6.3 Definition

6.3.1 Scale error. Scale error is the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure.

6.4 External connections. When external connections to simulate installed conditions are required, they will consist of 10 feet of 1/4-inch copper or aluminum tubing so coiled that moisture can drain out the open end. Similar coils should be connected to the pitot and static pressure connections.

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6.5 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force -11

Preparing activity:
Air Force - 11

Review activity:
Air Force - 71

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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